1. (a) Index number = 1893D = 765H

So, Local table selected. TI = 1.

Selector = 0011 1011 0010 1 | 1 | 10

765 | TI | Given RPL

= 3B2EH (Ans.)

(b) For 765H, Physical Address: 4E3483B1H

Base ~~Offset~~ Address: Physical Address − Offset

= 4E3483B1H − 5E9A1H

= 4E2E9AJ0H (Ans.)

Ending Address: Base + (limit × 1K)

= 4E2E9A10H + (1B3C6H × 400H) + 3FFH

= 52FDB60FH. (Ans)

(c) Access Right: B7H = ~~000~~ 1011 0111

P | DPL | E | A

Yes. A = 1; Segment is accessed.

(d) E = 0; So, Data or Stack. ED = 1. [Downward].

So, Stack Segment (SS)

(e) P = 1. So, Defined.

(f) DPL = 01.

2. a) let, n-bit ADC is used. So,

$$\frac{2^n - 1}{15} = \frac{10922}{10} \Rightarrow 2^n - 1 = 16383 \Rightarrow 2^n = 2^{14} \therefore n = 14$$

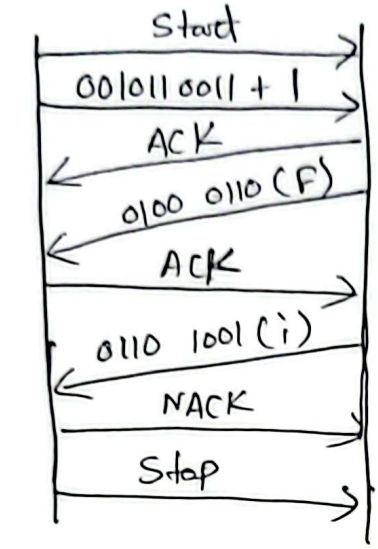
b) $\lceil \log_2(15000) \rceil = 14$

c) RISC Architecture.

i. Fixed-length ins.

ii. Large number of registers.

iii. Simple ins.

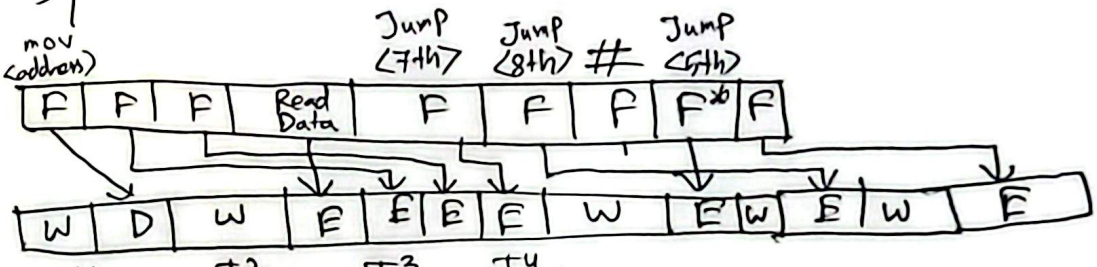


Slave-3
= B3 H
= 00 1011 0011

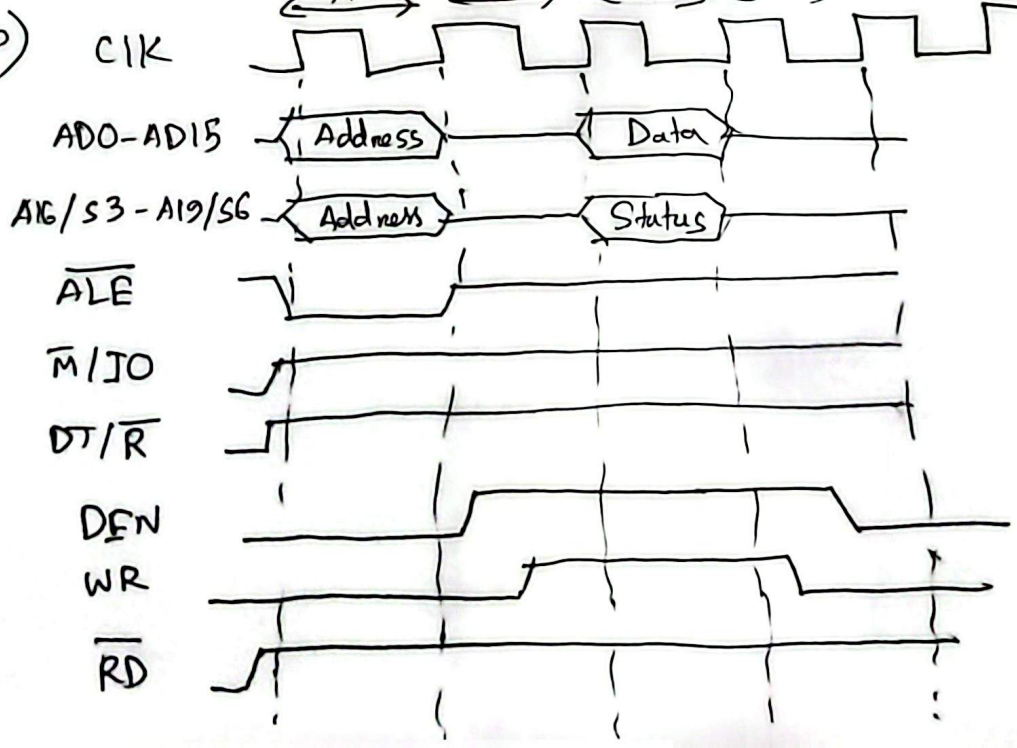
F (70)
= 0100 0110

i (105)
= 0110 1001

3. a)



b)



CamScanner

4 a) Given,

$7 * 2^2 - 9$

$= 7 * 2 * 2 - 9$

$= (7+7) * 2 - 9$

$= (7+7+7+7) - 9$

Assembly:

```
LDA  8H
ADD  8H
ADD  8H
ADD  8H
SUB  9H
OUT
HLT
```

b) $CON = C_p \overline{E_u}\, \overline{L_M}\, S_u \quad L_A \overline{CE}\, \overline{L_I}\, E_I \quad \overline{L_B}\, \overline{E_P}\, L_o E_A$

Disable: 0 1 1 0 0 1 1 0 1 1 0 0

$= 66CH$

i) $CON = 2CCH$; Acc = 2EH, B = 1EH, IR = 1FH, MAR = 6H

$= 0\ \boxed{0}\ 1\ 0 \quad \boxed{1}\ 1\ \boxed{0}\ 0 \quad 1\ 1\ 0\ 0$

($\overline{E_u}$, L_A, $\overline{L_I}$)

Adder $\xrightarrow{WBUS}$ Acc, IR

$\therefore$ **Acc = 2E + 1E = 4CH**, **IR = 4C**, **others remain same**

ii) CON = 47CH

= 01 [0] 0 011 [1] 1100

L_M E_I

IR $\xrightarrow{WBUS}$ MAR

∴ MAR = CH (Only lower nibble (4 bit) is received); ~~others~~

Others remain same.

iii) CON = 624H

= 01 10 0 [0] 10 [0] 1 00

CE L_B

RAM[MAR] $\xrightarrow{WBUS}$ B Reg

Since, MAR = CH ; ~~RAM[CH] =~~ RAM[CH] = 3BH

∴ B Reg = 3BH ; others remain same.

iv) CON = 364H

= 0 [0] 1 [1] 0110 [0] 100

E_u S_u L_B

Subtractor $\xrightarrow{WBUS}$ B Reg

∴ B Reg = 4C − 3B = 11H ; others remain same.

~~After~~	Acc = 2EH	B Reg = 1EH	Output = 2DH
T3	4C	1E	2D
T4	4C	1E	2D
T5	4C	3B	2D
T6	4C	11	2D

4) c) i) SUBOUT XH

T5: RAM $\xrightarrow{WBUS}$ B Reg

RAM ↙ C_E ; B ↓ L_B

CON = 0110 0010 0100

= 624H

T6: Adder/Subtractor $\xrightarrow{WBUS}$ OUT

Adder/Subtractor ↙ S_u, E_u ; OUT ↓ L_o

CON = 0011 0110 1110

= 36EH

ii) OUTM XH:

T4: IR $\xrightarrow{WBUS}$ MAR

IR ↙ E_I ; MAR ↓ L_M

CON = 0100 0111 1100

= 47CH

T5: RAM $\xrightarrow{WBUS}$ OUT

RAM ↙ C_E ; OUT ↓ L_o

CON = 0110 0010 1110

= 62EH

T6: Disable

CON = 66CH